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ABSTRACT

The alignment of Raman beams is important for high-precision atom interferometry gravimeters, and tilt modulation experiments are usually carried out to calibrate the direction of Raman beams. However, the Coriolis effect could be changed during the tilt modulation experiment, thus introducing a deviation in the alignment of Raman beams. In this work, a method of rotating the sensor head by 90° to eliminate the impact of the Coriolis effect on Raman beams' alignment in a fountain atom gravimeter was proposed and demonstrated experimentally. Using this method, the alignment uncertainty of the Raman beams in the atom gravimeter was reduced to 11 μrad, which contributed an uncertainty of less than 0.1 μGal (1 μGal = 1×10^{-8} m/s^2) to the gravity measurement. Further simulations explained the experimental results and pointed out a way to reduce the Coriolis effect in atom gravimeters.

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I. INTRODUCTION

Gravimetry based on atom interferometry[1] has experienced rapid development[2–6] and has been widely applied[7–11] in recent years. State-of-the-art atom gravimeters have achieved a resolution at the sub-μGal level.[4–6,12,13] However, the uncertainty of these gravimeters has not yet been reduced below 1 μGal, primarily due to the effect of wavefront aberrations.[6,12–14] In addition, other effects, such as the tilt of Raman beams, must be prioritized in high-precision atom gravimeters. The tilt of Raman beams can introduce significant errors if not systematically studied. Therefore, the accurate calibration of this tilt is a critical issue in the development of atom gravimeters.

The phase of an atom interferometer can be expressed as

$$\Delta\phi = \vec{k}_{eff} \cdot \vec{g} T^2, \tag{1}$$

where $\vec{k}_{eff}$ is the effective wave vector of the Raman beams, $\vec{g}$ is the gravitational acceleration, and T is the time interval between two Raman pulses. When there is an angle θ between the Raman beams and the direction of gravity, the phase can be expressed as $\Delta\phi = k_{eff} g T^2 \cos\theta$. Consequently, the systematic error induced by the tilt of the Raman beams is $\Delta g_t = g(\cos\theta - 1) \approx -\frac{1}{2} g\theta^2$ when θ is much smaller than 1 rad. Thus, the vertical alignment of the Raman beams can be determined by modulating the angle θ between $\vec{k}_{eff}$ and $\vec{g}$. This modulation experiment can be decomposed into two independent experiments along the x and y directions.

However, in certain experimental configurations, modulating the Raman beam direction may alter other system parameters. In our atom gravimeter, the quarter-wave plate (QWP) and retro-reflecting mirror are installed inside the vacuum chamber to minimize the effect of wavefront aberrations caused by the bottom window of the vacuum chamber. Consequently, Raman beam direction modulation requires rotating the entire vacuum chamber, which induces the velocity variations of the atomic cloud perpendicular to the Raman beam direction. However, the Coriolis effect caused by the Earth's rotation induces velocity dependent phase shifts in an atom interferometer, as illustrated in Fig. 1(c). Developing effective methods to mitigate the Coriolis effect during tilt modulation is,

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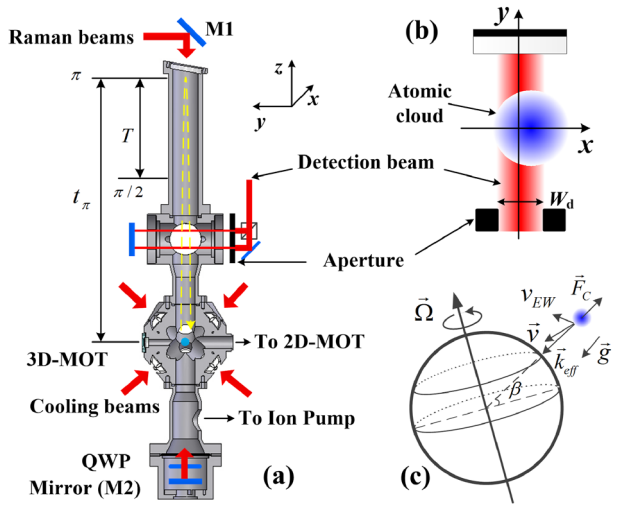


FIG. 1. Physical system of our atom interferometry gravimeter. (a) The physical system. (b) Top view of the detection system. The width of the detection beam in the *x* direction is 11 mm, obtained by selecting the central 11 mm portion from a Gaussian beam with a diameter of 22 mm, while its width in the *z* direction is 5 mm. T is interval time between two Raman pulses, and t_π is the interval time between launch and π pulse. (c) Schematic diagram of the Coriolis effect.

therefore, crucial for precise Raman beams' alignment in such gravimeters.

The phase shift induced by the Coriolis effect in atom gravimeters depends on Earth's rotation rate and the atomic cloud's average velocity in the EW direction, which can be expressed as[15]

$$\Delta\phi_\Omega = 2\vec{k}_{eff} \cdot (\vec{\Omega} \times \vec{v}) T^2, \tag{2}$$

where $\vec{\Omega}$ is the rotation rate of Earth and $\vec{v}$ is the velocity of the atomic cloud. The bias induced by the Coriolis effect in an atom gravimeter is $\Delta g_\Omega = 2\Omega v_{EW} \cos\beta$, where v_{EW} is the velocity of the atomic cloud in East–West direction and β is the latitude of our laboratory. Two conventional compensation methods exist: (1) gravimeter reversal[12,14] and (2) retro-reflecting mirror rotation for Earth rotation compensation.[15] The first method is exclusively suitable for evaluating the Coriolis effect, while the second method can effectively eliminate the Coriolis effect in tilt modulation experiments. However, the second approach presents implementation challenges for systems where the retro-reflecting mirror is mounted in vacuum.

This work presents an alternative approach involving 90° sensor head rotation to eliminate the Coriolis effect during Raman beam alignment. Our method achieved Raman beam direction calibration with an uncertainty of 11 μrad, contributing just 0.06 μGal to the gravitational measurement uncertainty. Furthermore, a theoretical analysis was conducted to demonstrate that the Coriolis effect can be effectively mitigated by restricting the probe beam width along the east–west direction, thereby providing a plausible explanation for the phenomena observed in the present experiments.

II. PROPOSAL TO ELIMINATE THE CORIOLIS EFFECT IN TILT MODULATION EXPERIMENTS

Experiments carried out in this work are based on a fountain atom interferometry gravimeter, which is mainly composed of two parts, namely a physical system and a laser system. The laser system is described in detail in our former work.[16] The physical system is displayed in Fig. 1(a). Usually, in an atom gravimeter, the QWP and retro-reflecting mirror are mounted outside the vacuum chamber. However, in this system, they are mounted inside the vacuum chamber to avoid wavefront distortion induced by the bottom window, which usually contributes the largest uncertainty to the effect of wavefront aberrations.[17] This configuration requires the alignment of the Raman beams to be accomplished by two steps. First, to ensure spatial overlap between the incident and reflected Raman beams, the mirror M1, which is positioned above the vacuum chamber, is precisely aligned, as illustrated in Fig. 1(a). The mirror orientation is optimized to maximize the optical power of the reflected Raman beam coupled back into the fiber, thereby ensuring optimal beam coincidence. Then, to make the direction of $\vec{k}_{eff}$ coincident with the direction of $\vec{g}$, we adjust the tilt of whole vacuum chamber.

When the tilt modulation experiment is carried out in the north–south (NS) direction, the variation of gravity can be expressed as

$$\Delta g \approx -\frac{1}{2} g\theta^2. \tag{3}$$

However, when the tilt modulation experiment is carried out in the EW direction, the impact of Coriolis effect cannot be ignored, so the variation of gravity can be expressed as[18]

$$\Delta g \approx -\frac{1}{2} g\theta^2 - 2\Omega \cos\beta g t_\pi \theta, \tag{4}$$

where g is the gravitational acceleration and t_π is the interval time between launching and the Raman π pulse. A comparison between Eqs. (3) and (4) reveals a 37 μrad difference in the symmetry axes of the two parabolas. This indicates that when conducting tilt modulation experiments in the EW direction, neglecting the Coriolis effect would introduce a bias of −0.7 μGal in the gravity measurement. In the absence of detection aperture, the influence of Eq. (4) is shown by the black solid line in Fig. 4(a).

Thus, how to eliminate the impact of the Coriolis effect on Raman beams' alignment is important in this atom gravimeter. The Coriolis effect is sensitive to the average velocity in the EW direction of the atomic cloud, so the impact of the Coriolis effect will be introduced when the tilt modulation experiment is carried out in the EW direction. Therefore, if the tilt modulation experiments are only carried out in the NS direction, the impact of the Coriolis effect can be eliminated. Therefore, we firstly calibrate the *y* direction of the Raman beams, which is coincident with the NS direction, so this experiment will not include the impact of the Coriolis effect, as shown in Fig. 2(a). Then, the tilt modulation experiment in *x* direction is carried out after the sensor is rotated an angle of 90°, as shown in Fig. 2(b). In this situation, the impact of the Coriolis effect on the result of *x* direction is also eliminated. After the above steps, both directions of the Raman laser are aligned without the influence of the Coriolis effect.

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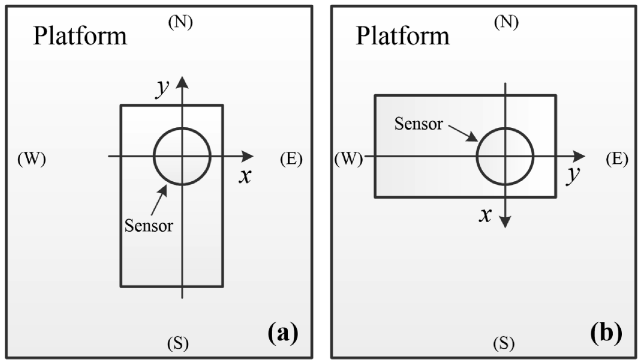


FIG. 2. Experimental strategy to eliminate the impact of the Coriolis effect in tilt modulation experiments. In position (a), the *y* direction of Raman beams is calibrated. In position (b), the *x* direction of Raman beams is calibrated.

III. EXPERIMENTAL DEMONSTRATION

A. Experimental results

A compass with an accuracy of 0.5° was used to determine the NS direction, and a reference line was drawn on the ground along this orientation. Although the compass points to the magnetic north, which deviates somewhat from Earth's rotational axis, it introduces an estimated bias of ~10%, which is smaller than the experimental uncertainty in this work; thus, this deviation does not affect the conclusions of this study. In the evaluation of the Coriolis effect, there is no specific requirement regarding the orientation of the instrument placement. It is only necessary to ensure that the rotation angle is strictly 180° in each trial. The gravimeter sensor cart was then aligned with its long side parallel to the reference line, corresponding to the position shown in Fig. 2(a). To demonstrate the proposal experimentally, tilt modulation experiments were firstly carried out in position (a). In this situation, the *y* direction of Raman beams can be calibrated. Then, we rotated the sensor with an angle of 90°, and the same modulation experiment was carried out, as shown in Fig. 2(b). A tiltmeter (Jewell 755) is employed to record the variation of tilt angle. The direction of the Raman beams is usually not coincident with the direction of gravity when the tiltmeter reading is zero. Therefore, the result of the tilt modulation experiment can be fitted to a parabolic function $y = a(\theta - \theta_0)^2 + b$, where θ_0 represents the tiltmeter reading when the direction of the Raman beams coincides with the direction of gravity.

The experimental results of tilt modulation in the *y* direction are displayed in Fig. 3(a), where the blue circles represent the results in position (a) and the red squares represent the results in position (b). The uncertainty associated with each experimental data point in the figure ranged from 1 to 2 μGal, as illustrated in the inset of Fig. 3(b). The angle difference between two positions was 28(5) μrad, which was slightly different from the calculation. The result has been confirmed by the same experiment five times, and the average difference between these two configurations was 27(12) μrad, corresponding to the difference between the values in the third and fourth rows of Table I, which was consistent with the theoretical value within the uncertainty range. The observed deviation may originate from the misalignment of the *y* direction relative to the strict north–south or east–west orientation.

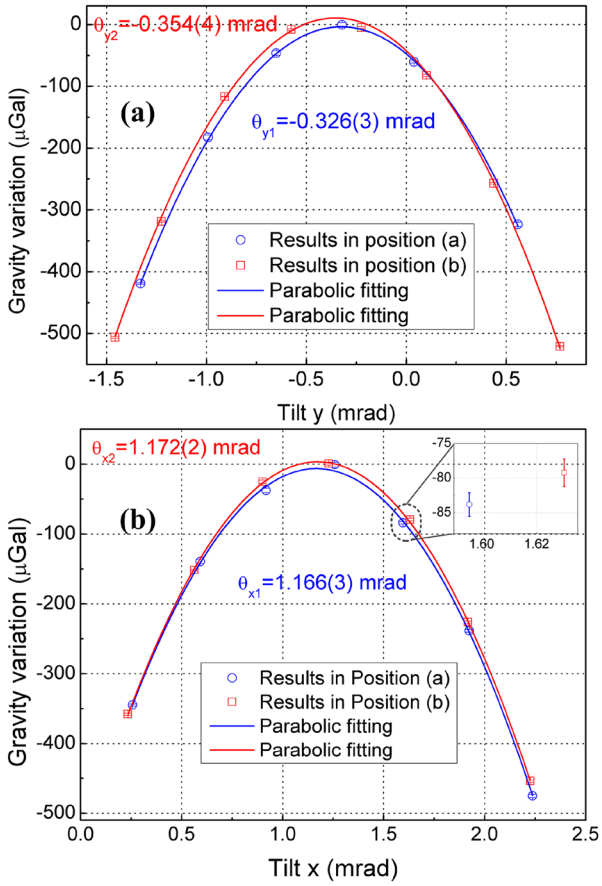


FIG. 3. Results of tilt modulation experiments. The tilt modulation results of *y* direction are shown in (a), and the results of *x* direction are displayed in (b).

TABLE I. Axes of symmetry in different configurations.

Axes of symmetry	Values (mrad)	Uncertainties (mrad)
X in configuration (a)	1.172	0.011
X in configuration (b)	1.174	0.005
Y in configuration (a)	−0.327	0.009
Y in configuration (b)	−0.354	0.008

However, there was an obvious deviation between experimental result and theoretical prediction when the tilt modulation experiment was carried out in the *x* direction. When the tilt modulation experiment of the *x* direction was carried out in position (a), results are displayed as blue spheres in Fig. 3(b). The experimental results in position (b) are displayed as red squares in Fig. 3(b). The fitting results indicated that the angle difference between two positions

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was only 6(4) μrad, which was obviously different from the theoretical result (37 μrad). These experiments have been carried out three times, and similar results were obtained. The average of the three measurements was 2(12) μrad, corresponding to the difference between the second and first rows of Table I. How to explain this result will be given in Sec. III B.

The tilt modulation experiment conducted along the NS direction was used as the basis for calibrating the Raman beam orientation, corresponding to the values listed in the second and third rows of Table I; thus, the total uncertainty in the Raman beam alignment was determined to be $\delta\theta = \sqrt{\delta\theta_x^2 + \delta\theta_y^2} = \sqrt{0.005^2 + 0.009^2} = 0.011$ mrad. This contributed merely $\delta g = 0.5 g_0 \delta\theta^2 = 0.06$ μGal to the overall uncertainty in gravity measurement.

B. Reducing the impact of Coriolis effect by limiting the width of detection beam

After considering multiple factors, such as misalignment with the NS direction and variations in tiltmeter readings, we concluded that the width limitation of the probe beam is the dominant factor. To homogeneously detect all atoms after interferometry, the detection beam is usually designed with large width. In this situation, the Coriolis effect is linearly dependent on the average velocity of all atoms in the EW direction. However, if there is an aperture, which limits the width of detection beam in the EW direction, the Coriolis effect is dependent on the average velocity of detected atoms in the EW direction. In this situation, the average velocity of detected atoms is smaller than the average velocity of all atoms in the EW direction, so the Coriolis effect will be reduced.

A Monte Carlo method has been used to demonstrate this idea. At the initial moment, an atomic cloud is generated with both its position and velocity following Gaussian distributions. After a period of free flight, the spatial distribution of the atomic cloud can be described by

$$n(x,y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-x_t)^2}{2\sigma^2}\right] \exp\left[-\frac{(y-y_t)^2}{2\sigma^2}\right], \quad (5)$$

where $\sigma^2 = \sigma_0^2 + \sigma_v^2 t^2$ is the size of the atomic cloud at the moment of detection, σ_0 is the initial size of the atomic cloud, t is the total free-falling time of the atoms, $\sigma_v^2 = \frac{k_B T_a}{m}$ is the width of the velocity distribution of the atomic cloud, and T_a is the temperature of the atomic cloud. Furthermore, the central positions of the atomic cloud $x_t = x_0 + v_x t$ and $y_t = y_0 + v_y t$ can be determined by the initial positions and velocities of atomic cloud. The parameters related to the experiment, such as the initial position,[19] horizontal velocity,[19] and temperature of the atomic cloud,[20] are all obtained from experimental measurements.

Among the atoms that fall back into the detection region, only those within the illumination range of the detection beam can be detected. In particular, atoms whose x coordinate lies within $[-W_d/2, W_d/2]$ are considered detectable, where W_d represents the width of the detection beam along the x direction, as illustrated in Fig. 1(b). It should be noted that at this moment, the position and velocity of the atoms are correlated. By excluding all atoms outside the range of the detection beam and averaging the velocities of the remaining atoms, the average velocity of the atomic cloud is obtained. Subsequently, the magnitude of the Coriolis effect can be calculated by

$$\Delta g_\Omega = 2\Omega \cos\beta \sum_{i=1}^{N} v_{xi}/N, \quad (6)$$

where v_{xi} is the velocity of ith atom detected and N is the total number of detected atoms. The key parameters used in the simulation are listed in Table II, and the simulation results are displayed in Fig. 4.

From Fig. 4(a), it can be observed that the slope representing the change in the average velocity of detected atoms along the x direction with respect to tilt in the EW direction is smaller when a limitation is present compared to the case without limitation. It means that the Coriolis effect decreases with decreasing detection beam width, which has been demonstrated by experiments.[21] Due to the horizontal velocity of the atomic cloud, when the probe beam width is too small, the number of atoms that can be detected sharply decreases. To ensure that we can get enough time-of-flight (TOF) signal, the detection beam should not be designed with very small width. So, the width of detection beam in the x direction in our atom interferometer is 11 mm. With these parameters, the average velocity of detected atoms has been reduced 5.6 times compared to the average velocity of all atoms in the x direction with typical parameters. The result means that the Coriolis effect is also reduced 5.6 times when there is a limitation with a width of 11 mm for detection beam.

The total gravity bias from the EW tilt and the Coriolis effect, for a system with an 11 mm x direction aperture, is shown by the blue dashed line in Fig. 4(b). The gravity variation only induced by the tilt of the Raman beams is shown as red line in Fig. 4(b). The symmetry axis difference between these two situations is only 5.6 μrad, which is coincident with the experimental result in Fig. 3(b). Of course, this difference is negligible for Raman beams' alignment in micro-Gal level atom gravimeters. Therefore, the experimental results of tilt modulation in the x direction can be explained by these simulations.

TABLE II. Key parameters used in the calculation. The moment of Raman π pulse and interval time of Raman pulses are the experimental parameters related to our experimental configuration. Initial positions, horizontal velocities, and temperature of the atomic cloud are measured online.

Parameters	Symbols	Values
Rotation rate of Earth	Ω	72.92 μrad/s
Latitude of our lab	β	30°
Gravitational acceleration	g	9.79 m/s^2
Moment of Raman π pulse	t_π	0.297 s
Interval time between Raman pulses	T	0.16 s
Total free-falling time	t	0.47 s
Initial size of the atomic cloud	σ_0	1 mm
Initial x position of the atomic cloud	x_0	0.2 mm
Initial y position of the atomic cloud	y_0	−0.2 mm
Velocity in x direction	v_x	2.2 mm/s
Velocity in y direction	v_y	4.2 mm/s
Temperature of atomic cloud	T_a	3 μK

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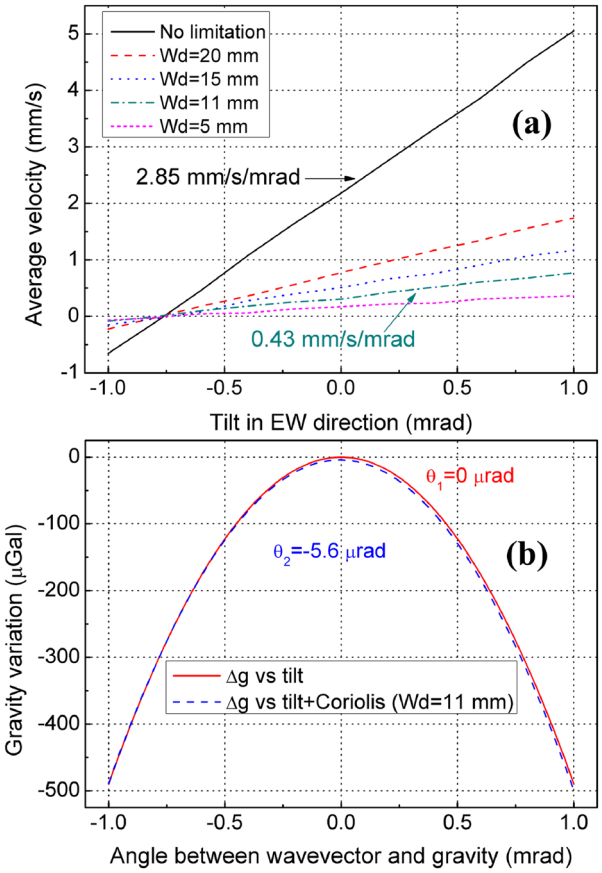


FIG. 4. Simulation results: (a) The average velocity of detected atoms in the *x* direction changes with the tilt of the EW direction with different width of detection beam. (b) The gravity changes with the tilt of the EW direction. Red line: the gravity variation is only induced by tilt. Blue dash: the gravity variation is induced by tilt and Coriolis effect when there is an aperture with a width of 11 mm in the EW direction.

IV. CONCLUSIONS AND DISCUSSIONS

In an atom gravimeter, the impact of the Coriolis effect is usually not considered when calibrating the direction of the Raman beam. However, the Coriolis effect depends on the tilt angle of the Raman beams. As a result, the vertical direction identified through tilt modulation experiments does not align with true vertical, introducing a systematic error in the atom gravimeter. In this work, the alignment deviation of the Raman beam caused by the Coriolis effect in a fountain atom gravimeter was analyzed first; then, a strategy to eliminate the impact of the Coriolis effect on Raman beam alignment was proposed and demonstrated. Using this method, the direction of the Raman beams has been aligned within an uncertainty of 11 μrad, contributing an uncertainty of 0.06 μGal to the gravity measurement, which is already far smaller than the measurement uncertainty of the most accurate atom gravimeters currently available.[12–14,20] In addition, the experimental results indicated that the influence of the Coriolis effect will be greatly suppressed when the tilt modulation experiment is carried out in the direction where the detection beam is limited. Further simulations validated the experiment and provided the suppression factor of the Coriolis effect when the detection beam is limited.

This method offers a way to eliminate the Coriolis effect's influence on Raman laser alignment without additional hardware. However, for systems already incorporating a tip-tilt stage, this compensation is inherently provided, making the method redundant for this specific purpose. In addition, the response of horizontal atomic velocity to platform tilt in free-falling atom interferometers differs from that in fountain configurations. Therefore, the specific impact of the Coriolis effect on alignment may also vary. Despite this, performing tilt modulation solely in the NS direction remains a reliable strategy to decouple and eliminate the Coriolis effect. This approach provides a basis for Raman beam calibration in miniaturized atom gravimeters and offers valuable guidance for the design of high-precision atom gravimeters.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Qin Luo: Conceptualization (equal); Data curation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Hang Zhou**: Data curation (equal); Formal analysis (equal); Software (equal); Writing – review & editing (equal). **Li Li**: Data curation (supporting); Formal analysis (supporting). **Lele Chen**: Data curation (supporting); Investigation (supporting); Methodology (supporting). **Xiaochun Duan**: Methodology (supporting); Supervision (supporting). **Zhongkun Hu**: Funding acquisition (supporting); Project administration (supporting). **Minkang Zhou**: Funding acquisition (lead); Project administration (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1] M. Kasevich and S. Chu, "Atomic interferometry using stimulated Raman transitions," Phys. Rev. Lett. **67**, 181–184 (1991).
[2] A. Peters, K. Y. Chung, and S. Chu, "Measurement of gravitational acceleration by dropping atoms," Nature **400**, 849–852 (1999).
[3] H. Müller, S.-w. Chiow, S. Herrmann, S. Chu, and K. Y. Chung, "Atom-interferometry tests of the isotropy of post-Newtonian gravity," Phys. Rev. Lett. **100**, 031101 (2008).

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[4]Z. K. Hu, B. L. Sun, X. C. Duan, M. K. Zhou, L. L. Chen, S. Zhan, Q. Z. Zhang, and J. Luo, "Demonstration of an ultrahigh-sensitivity atom-interferometry absolute gravimeter," Phys. Rev. A **88**, 043610 (2013).
[5]P. Gillot, O. Francis, A. Landragin, F. Pereira Dos Santos, and S. Merlet, "Stability comparison of two absolute gravimeters: Optical versus atomic interferometers," Metrologia **51**, L15–L17 (2014).
[6]C. Freier, M. Hauth, V. Schkolnik, B. Leykauf, M. Schilling, H. Wziontek, H.-G. Scherneck, J. Müller, and A. Peters, "Mobile quantum gravity sensor with unprecedented stability," J. Phys.: Conf. Ser. **723**, 012050 (2016).
[7]R. Geiger, V. Ménoret, G. Stern, N. Zahzam, P. Cheinet, B. Battelier, A. Villing, F. Moron, M. Lours, Y. Bidel, A. Bresson, A. Landragin, and P. Bouyer, "Detecting inertial effects with airborne matter-wave interferometry," Nat. Commun. **2**, 474 (2011).
[8]Y. Bidel, N. Zahzam, C. Blanchard, A. Bonnin, M. Cadoret, A. Bresson, D. Rouxel, and M. F. Lequentrec-Lalancette, "Absolute marine gravimetry with matter-wave interferometry," Nat. Commun. **9**, 627 (2018).
[9]X. Wu, Z. Pagel, B. S. Malek, T. H. Nguyen, F. Zi, D. S. Scheirer, and H. Müller, "Gravity surveys using a mobile atom interferometer," Sci. Adv. **5**, eaax0800 (2019).
[10]S. Merlet, P. Gillot, B. Cheng, R. Karcher, A. Imanaliev, L. Timmen, and F. Pereira dos Santos, "Calibration of a superconducting gravimeter with an absolute atom gravimeter," J. Geod. **95**, 62 (2021).
[11]L. Antoni-Micollier, D. Carbone, V. Ménoret, J. Lautier-Gaud, T. King, F. Greco, A. Messina, D. Contrafatto, and B. Desruelle, "Detecting volcano-related underground mass changes with a quantum gravimeter," Geophys. Res. Lett. **49**, e2022GL097814, https://doi.org/10.1029/2022GL097814 (2022).
[12]S. K. Wang, Y. Zhao, W. Zhuang, T. C. Li, S. Q. Wu, J. Y. Feng, and C. J. Li, "Shift evaluation of the atomic gravimeter NIM-AGRb-1 and its comparison with FG5X," Metrologia **55**, 360–365 (2018).
[13]P. W. Huang, B. Tang, X. Chen, J. Q. Zhong, Z. Y. Xiong, L. Zhou, J. Wang, and M. S. Zhan, "Accuracy and stability evaluation of the ^{85}Rb atom gravimeter WAG-H5-1 at the 2017 international comparison of absolute gravimeters," Metrologia **56**, 045012 (2019).
[14]A. Louchet-Chauvet, T. Farah, Q. Bodart, A. Clairon, A. Landragin, S. Merlet, and F. P. D. Santos, "The influence of transverse motion within an atomic gravimeter," New J. Phys. **13**, 065025 (2011).
[15]S. Y. Lan, P. C. Kuan, B. Estey, P. Haslinger, and H. Müller, "Influence of the Coriolis force in atom interferometry," Phys. Rev. Lett. **108**, 090402 (2012).
[16]Q. Luo, H. Zhang, K. Zhang, X. C. Duan, Z. K. Hu, L. L. Chen, and M. K. Zhou, "A compact laser system for a portable atom interferometry gravimeter," Rev. Sci. Instrum. **90**, 043104 (2019).
[17]V. Schkolnik, B. Leykauf, M. Hauth, C. Freier, and A. Peters, "The effect of wavefront aberrations in atom interferometry," Appl. Phys. B **120**, 311–316 (2015).
[18]L. Li, L. L. Chen, Q. Luo, H. Zhou, M. K. Zhou, and Z. K. Hu, "Coriolis effect due to laser beam tilting in an atom interferometer," Opt. Express **33**, 462–474 (2025).
[19]Q. Luo, H. Zhou, L. Chen, X. Duan, Z. Hu, and M. Zhou, "Determining the three-dimensional position of an atomic cloud in an atom interferometer using one laser beam," Phys. Rev. Appl. **17**, 024033 (2022).
[20]Q. Luo, L. Li, H. Zhou, L. Chen, X. Duan, Z. Hu, and M. Zhou, "Evaluating the effect of wavefront aberrations in an atom gravimeter," Metrologia **62**, 055006 (2025).
[21]B. Wu, D. Zhu, B. Cheng, L. Wu, K. Wang, Z. Wang, Q. Shu, R. Li, H. Wang, X. Wang, and Q. Lin, "Dependence of the sensitivity on the orientation for a free-fall atom gravimeter," Opt. Express **27**, 11252–11263 (2019).

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